

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B. Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No :16

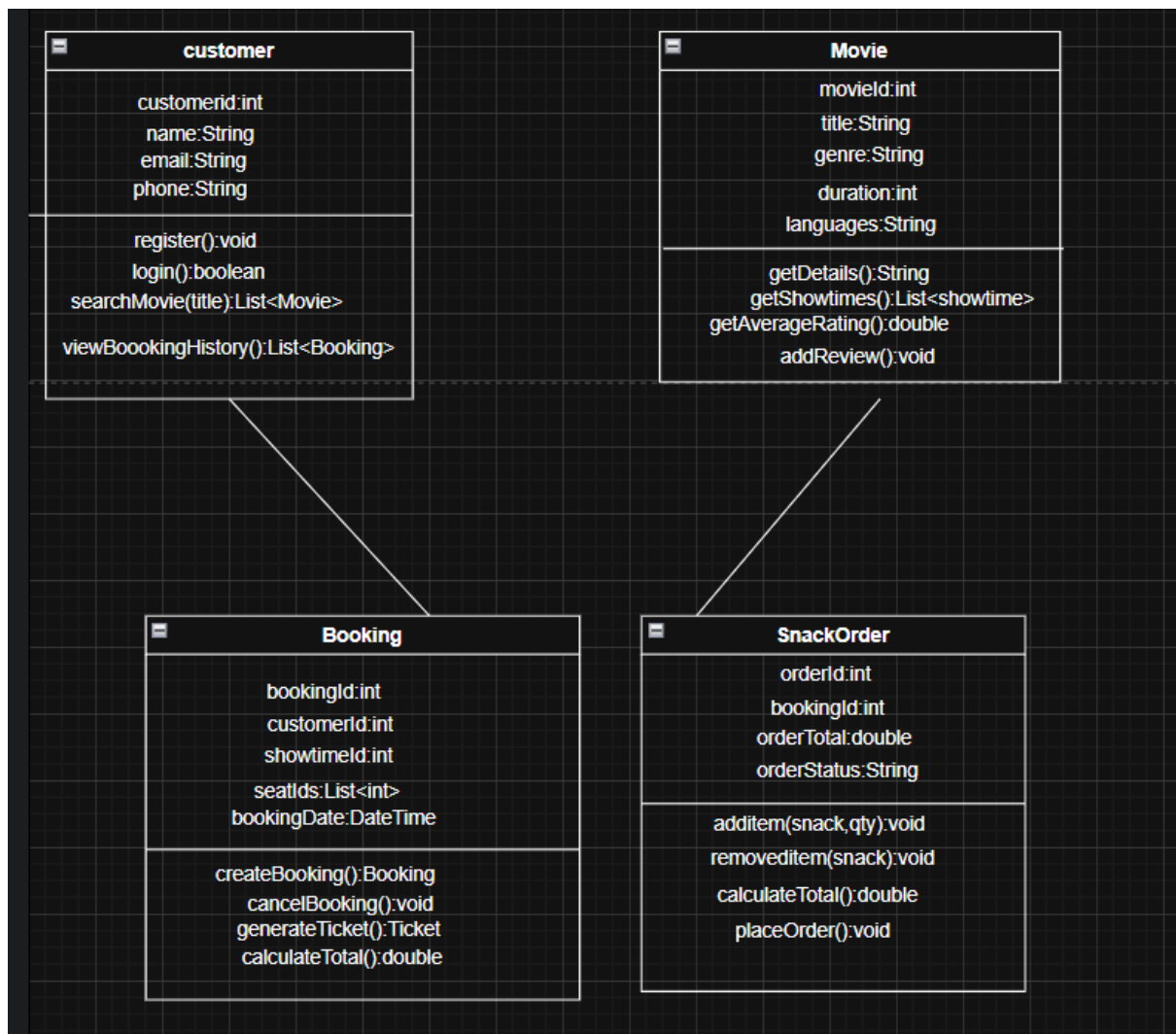
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1. Problem Title : Movie Ticket Booking & Snack Ordering System

STRUCTURAL DESIGN

2. Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)



BEHAVIUIORAL DESIGN

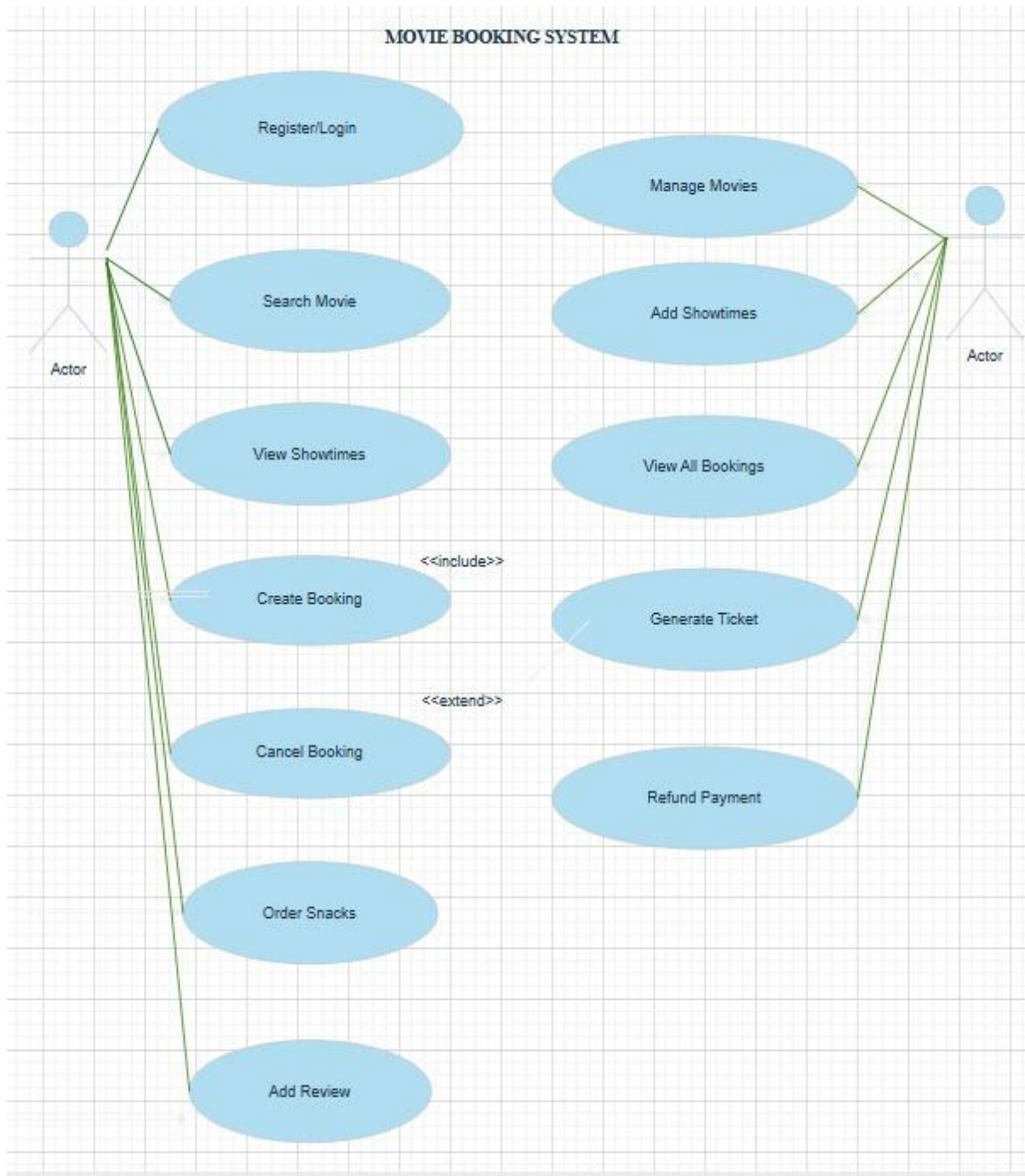
3. Use-case diagram

a. List of actors

- 1.Customer — registers, searches movies, books tickets, orders snacks, makes payment, cancels booking
2. Admin — manages movies, theatres, showtimes; views reports and analytics

b. List of use-cases

1. Register / Login — The customer creates a new account or logs in with existing credentials to access the system.
2. Search Movies — The customer searches for movies by title, genre, language, or date to find what is currently showing.
3. Select Showtime & Seats — The customer picks a preferred showtime and selects available seats from the seat map.
4. Add Snacks — The customer optionally browses the snack menu and adds items (popcorn, drinks, combos) to their order.
5. Make Payment — The customer pays for the booking (and snacks) through the integrated payment gateway.
6. Get Booking Confirmation — The system generates a booking ID and e-ticket/QR code sent to the customer's email.
7. Cancel Booking — The customer can cancel a confirmed booking and receive a refund as per the policy.
- 8/. View Reports (Admin) — Admin views ticket sales, occupancy, and revenue reports.



4. Sequence diagram (use UML notations)

a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

